21-242 Recitation 12

Subspaces — Q3. Without linearity, those sets need not be subspaces

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Show that if f is not a linear transformation, $\{\vec{v} \in V : f(\vec{v}) = \vec{0}\}$ and $\{f(\vec{v}) : \vec{v} \in V\}$ need not be subspaces.

Course-level solution

Take $V = W = \mathbb{R}$ over $\mathbb{R}$.

Zero-set need not be a subspace. Let $f(x) = x^2 - 1$. Then $\{x : f(x) = 0\} = \{-1, 1\}$, which does not contain 0 and is not closed under addition.

(Even simpler: $f(x) = x + 1$. Then $\{x : f(x) = 0\} = \{-1\}$, not a subspace.)

Image need not be a subspace. Let $g(x) = x^2$. Then $\{g(x) : x \in \mathbb{R}\} = [0, \infty)$, which is not closed under scalar multiplication by -1.

(Alternatively $g(x) = 1$ constantly: the image is $\{1\}$, which does not contain 0.)